When Sample Selection Bias Precipitates Model Collapse

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Introduction

Synthetic data now flows back into future training sets, so recursive training can amplify small distributional errors into model collapse (Shumailov et al., Nature 2024). Verification and sample selection are meant to stop this loop and avoid model collapse by keeping samples that match a trusted reference (Fu et al., NeurIPS 2025). However, in low-resource communities, for instance, data silos can make trusted references local and incomplete, so valid rare modes may look like bad samples and be filtered away.

Theory

Proposition 1: Replace paradigm (Shumailov et al., Nature 2024)

Under the *Replace* paradigm, as $t \to \infty$,

$$\boldsymbol{\Sigma}_t \xrightarrow{a.s.} \mathbf{0}, \qquad \mathbb{E}\left[\mathbb{W}_2^2(\mathcal{N}_t, \mathcal{N}_0)\right] \to \infty.$$

Synthetic-data replacement loops lose variance and drift from reality.

Proposition 2: Accumulate paradigm (Kazdan et al., ICML 2025)

Under unbiased accumulation,

$$\mathbb{E}[(\bar{\boldsymbol{\mu}}_t - \bar{\boldsymbol{\mu}}_0)^2] \to (1-\alpha_n)\bar{\boldsymbol{\Sigma}}_0, \quad \mathbb{E}[\bar{\boldsymbol{\Sigma}}_t] \to \alpha_n \bar{\boldsymbol{\Sigma}}_0.$$

Thus $\mathbb{E}[\mathbb{W}_2^2] \to 2(1-\sqrt{\alpha_n})\,\mathrm{Tr}(\bar{\boldsymbol{\Sigma}}_0)$, where $\alpha_n = \sin(\pi/\sqrt{n})/(\pi/\sqrt{n})$.

Unbiased accumulation can preserve diversity and slow collapse.

Theorem 1: local selection collapses diversity

With top-α selection toward a local ideal $\mathbf{u}^*$,

$$\|\bar{\boldsymbol{\mu}}_t - \mathbf{u}^*\|^2 \to 0, \quad \bar{\boldsymbol{\Sigma}}_t \to \mathbf{0}, \quad \mathbb{E}[\mathbb{W}_2^2] \to \|\mathbf{u}^* - \bar{\boldsymbol{\mu}}_0\|^2 + \mathrm{Tr}(\bar{\boldsymbol{\Sigma}}_0).$$

Local selection keeps familiar samples but erases global modes.

Theorem 2: diversity decays by power law

If the trajectory remains in the local basin,

$$\mathrm{Tr}(\bar{\boldsymbol{\Sigma}}_t) = \mathcal{O}_{a.s.}(t^{-\psi}).$$

Tail erosion has an explicit rate.

Once loops lock onto a local view, tail diversity decays predictably.

Theorem 3: drift controls true-manifold risk

For filtered distribution $\mathcal{D}_t$ and true manifold $\mathcal{D}^*$,

$$\mathcal{R}_{\mathcal{D}^*}(h_t; g^*) \le 2\ell\epsilon\,\mathbb{W}_p(\mathcal{D}_t, \mathcal{D}^*) + \mathcal{R}_{\mathcal{D}_t}(h_t; g_t) + \mathcal{O}(\ell\delta).$$

Drift from the true manifold controls downstream risk.

For more methodology and theoretical details, please refer to the full paper.

Synthetic data selection mechanisms in low-resource communities may bias recursive loops toward model collapse.

Scan

OpenReview paper and supplement.

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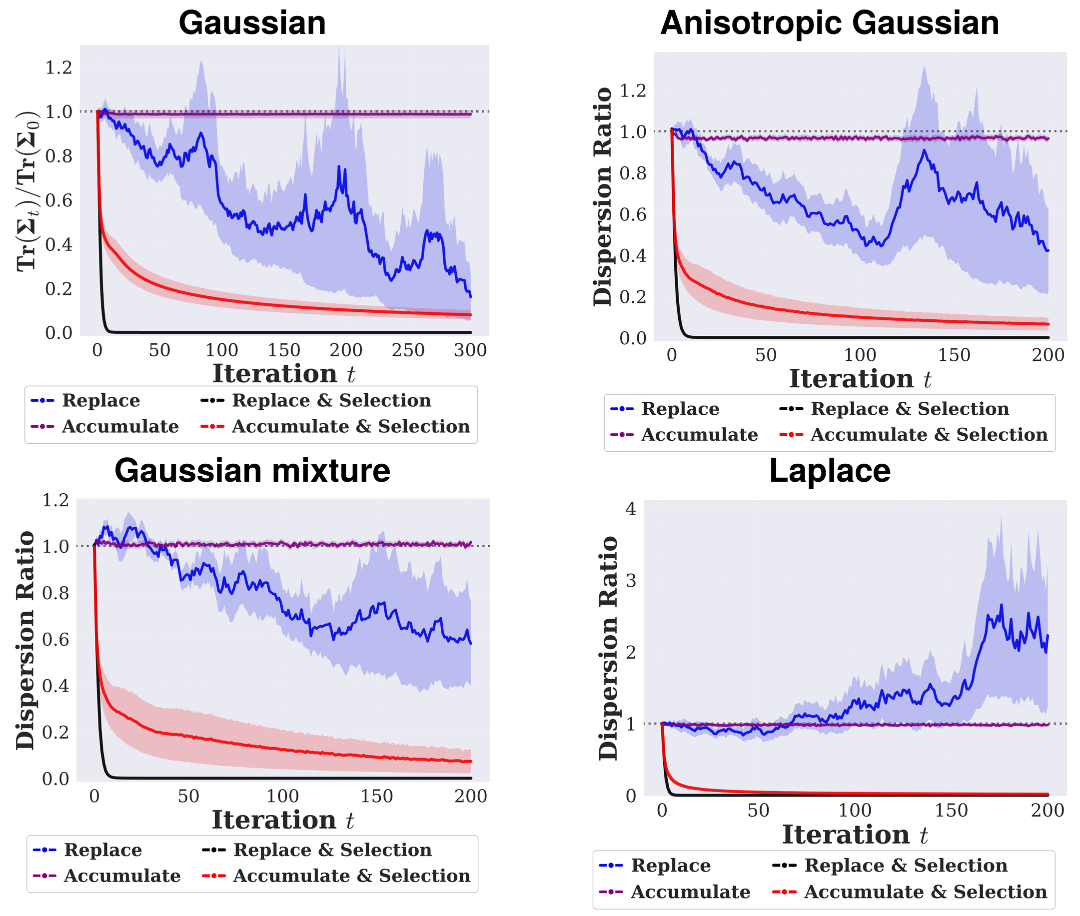
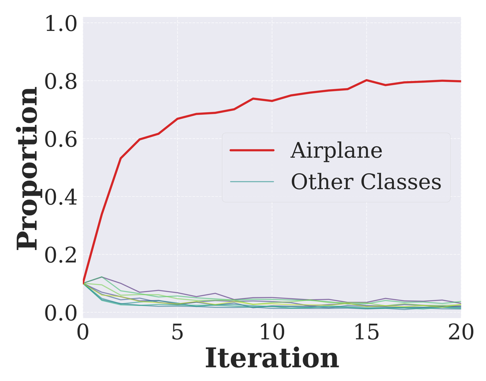
Distributional Modeling Results



Collapse Signal



Airplane-only verification moves CIFAR-10 toward Airplane and prunes other classes.

Experimental Takeaway

- Biased selection contracts synthetic distributions across modeling settings.
- Airplane-only verification concentrates CIFAR-10 on Airplane and prunes other classes.
- DDPM experiments in the full paper suggest non-IID references can make selection baselines lag Random, while broader references improve FID and recall.

References

[1] Shumailov et al. AI models collapse when trained on recursively generated data. Nature, 2024.

[2] Kazdan et al. Collapse or Thrive: Perils and Promises of Synthetic Data in a Self-Generating World. ICML, 2025.

[3] Fu et al. Self-Verification Provably Prevents Model Collapse in Recursive Synthetic Training. NeurIPS, 2025.

For additional experimental results, please refer to the full paper.

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